

DAY-5 DATA CLEANING

```
jupyter Data cleaning DAY-4,5 Last Checkpoint: yesterday

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JupyterLab Python [conda env:base] Trusted

[62]: #count duplicates:
import pandas as pd
print(df.duplicated().sum())
#since here we dont have any duplicates it prints as 0
0

[63]: #5. Dropping Duplicate Records
#before
print(df.shape)
#remove duplicates
df.drop_duplicates(inplace=True)
#after
print(df.shape)
#since there is no duplicates the shape vaal remains same
(52, 6)
(52, 6)

[64]: print(df.columns)
Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email'], dtype='object')

[65]: #6 string operations
#string methods work only on text columns
#remove Extra spaces
df['Student_Name'] = df['Student_Name'].str.strip()
#convert to uppercase
df['Student_Name'] = df['Student_Name'].str.upper()
print(df['Student_Name'].head())
#convert to title case
df['Student_Name'] = df['Student_Name'].str.title()
print(df['Student_Name'].head())
```

```
jupyter Data cleaning DAY-4,5 Last Checkpoint: yesterday

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JupyterLab Python [conda env:base] Trusted

0 Student1
1 Student2
2 Student3
3 Student4
4 Student5
Name: Student_Name, dtype: object
0 Student1
1 Student2
2 Student3
3 Student4
4 Student5
Name: Student_Name, dtype: object

[72]: df['Department'] = df['Department'].str.replace(
'CSE',
'Computer Science'
)
print(df['Department'].head())
# to replace single value str is used to replace multiple value str should not be used
#Replace Text
#Replace department name
df['Department'] = df['Department'].replace({
'CSE': 'Computer Science',
'ECE': 'Electronic and Communi eng',
'EEE': 'Electrical and Electronic Eng',
'IT': 'Information Tech',
'MECH': 'Mechanical Eng'
})
print(df['Department'].head())
0 Information tech
1 Electronics and communi eng
2 electrical and electronic
3 Mechanical
```

```
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JupyterLab Python [conda env:base] Trusted

3 Mechanical
4 Computer Science
Name: Department, dtype: object
0 Information tech
1 Electronics and communi eng
2 electrical and electronic
3 Mechanical
4 Computer Science
Name: Department, dtype: object

[73]: #String Length
df['Name_length'] = df['Student_Name'].str.len()
print(df[['Student_Name', 'Name_length']].head())

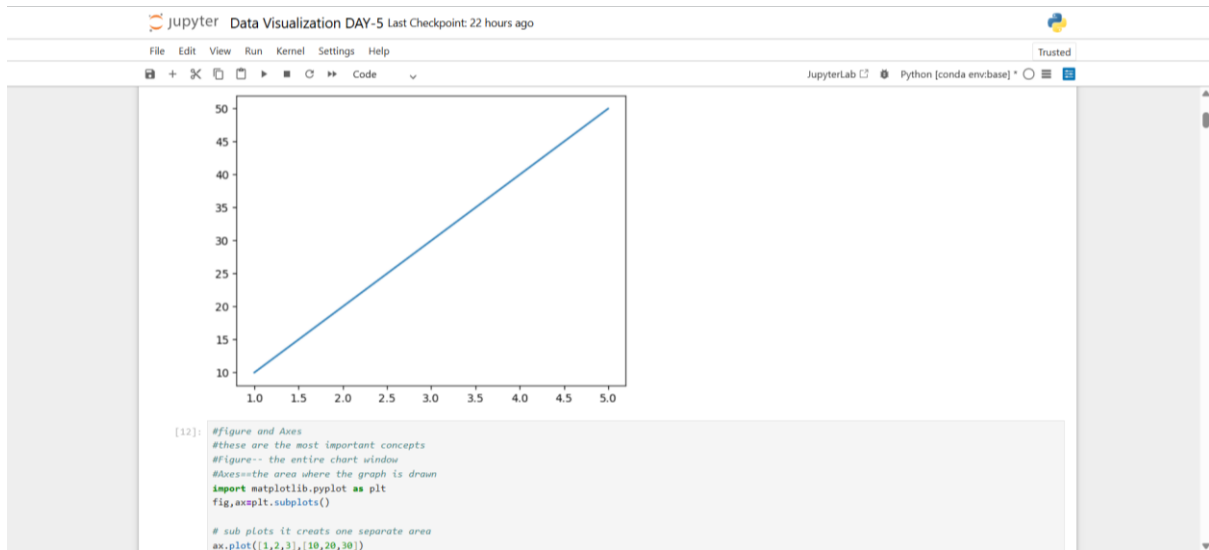
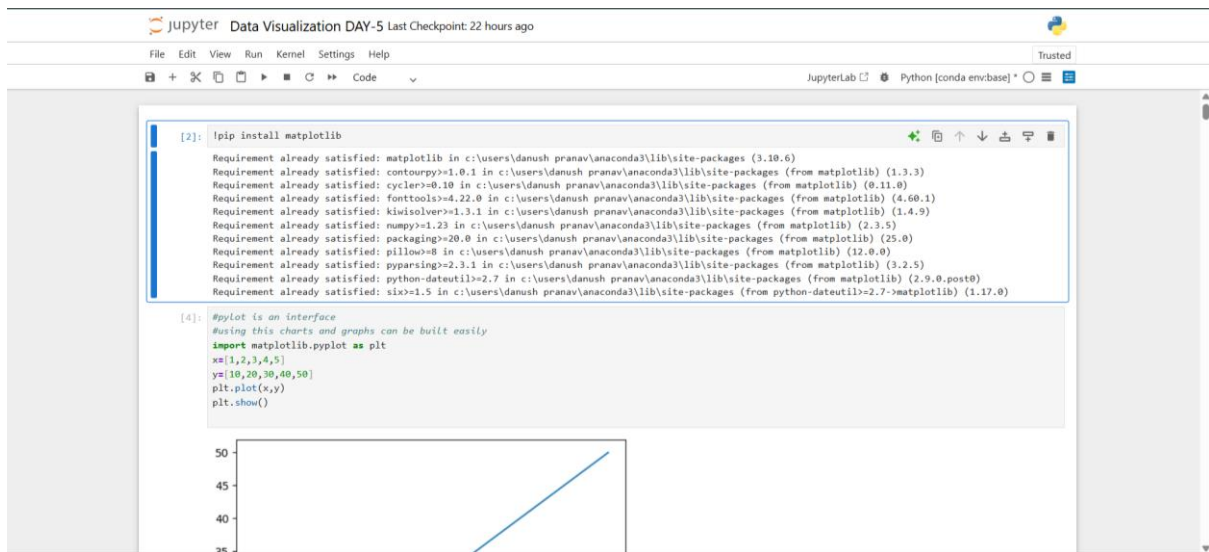
Student_Name Name_length
0 Student1 8
1 Student2 8
2 Student3 8
3 Student4 8
4 Student5 8

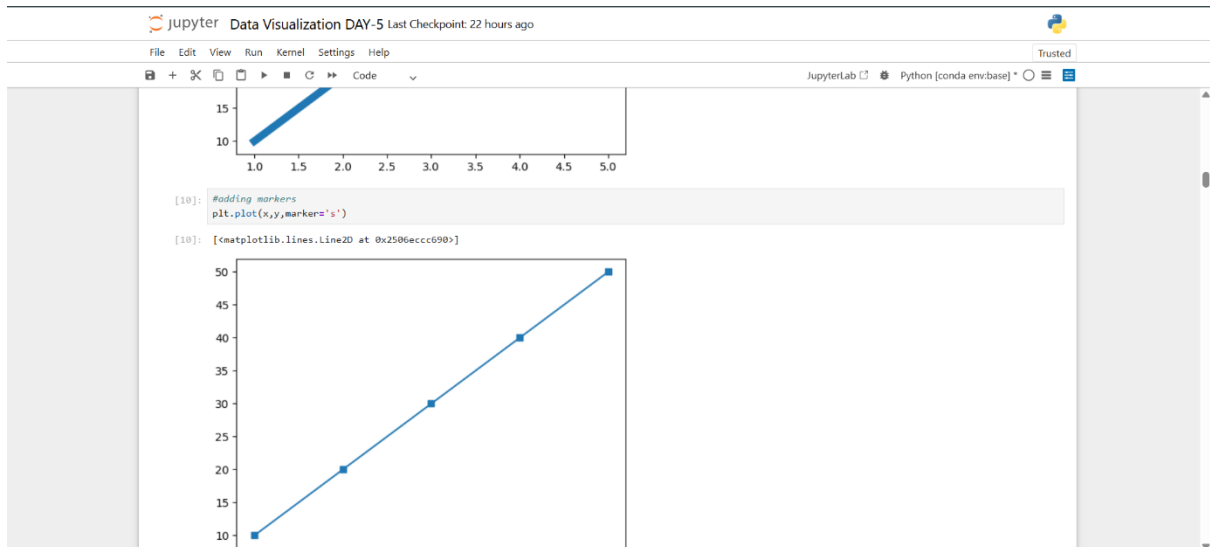
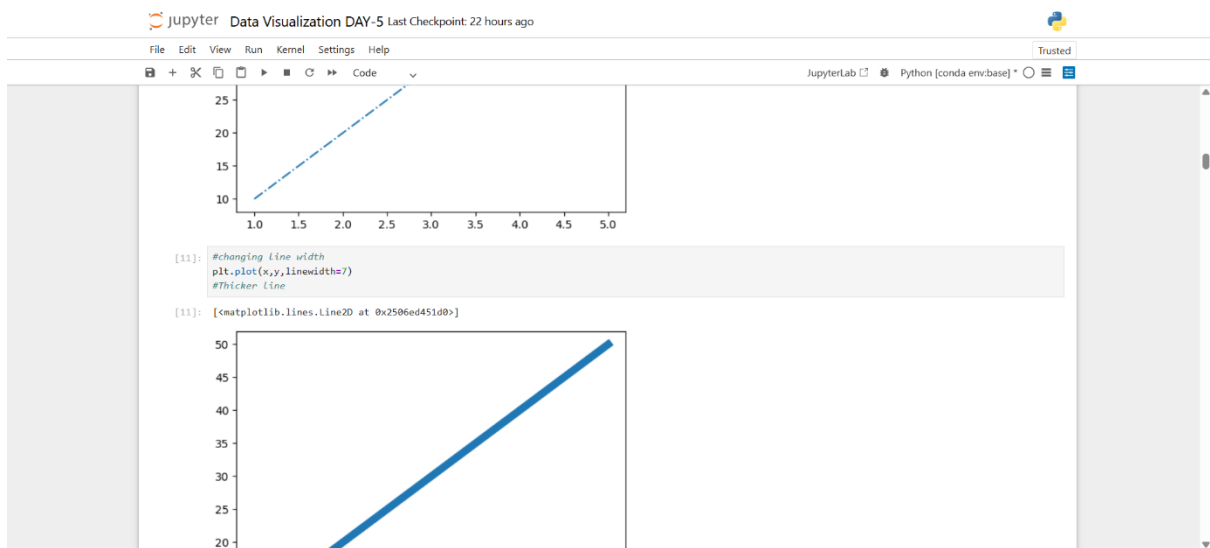
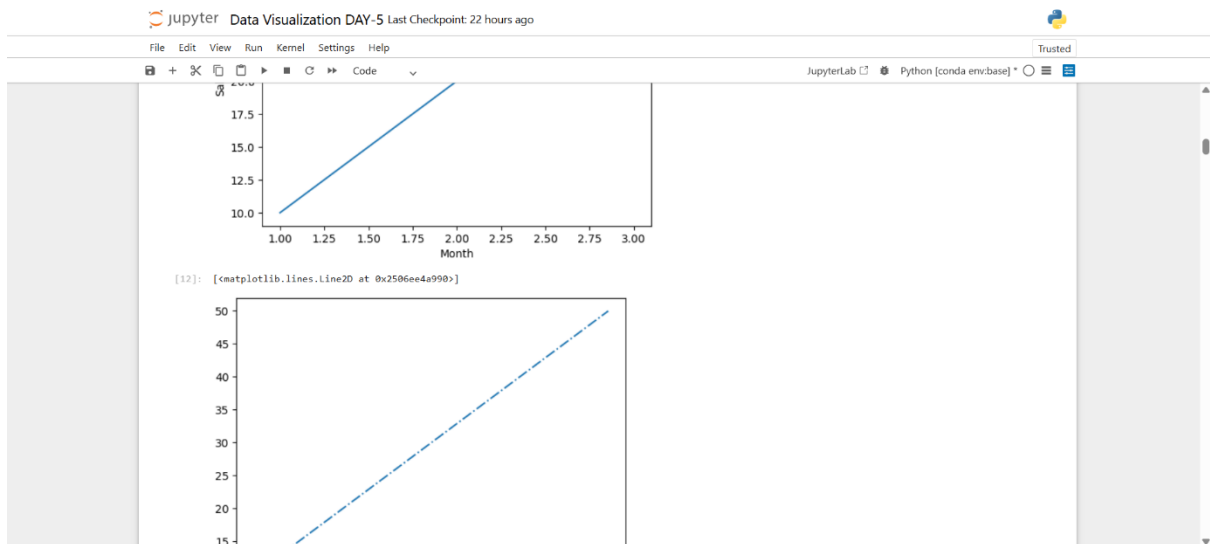
[74]: #Extract First character
df['First_Letter'] = df['Student_Name'].str[0]
print(df[['Student_Name', 'First_Letter']].head())

Student_Name First_Letter
0 Student1 S
1 Student2 S
2 Student3 S
3 Student4 S
4 Student5 S

[ ]:
```

DATA VISUALIZATION





Jupyter Data Visualization DAY-5 Last Checkpoint: 22 hours ago

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JupyterLab Python [conda env:base] * ○ □ ≡

```
[ ]: # 'o' Circle
# 's' Square
# '^' Triangle
# '*' Star

[13]: !pip install seaborn

Requirement already satisfied: seaborn in c:\users\danush prana\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy<1.24.0,>=1.20 in c:\users\danush prana\anaconda3\lib\site-packages (from seaborn) (2.3.5)
Requirement already satisfied: pandas>=1.2 in c:\users\danush prana\anaconda3\lib\site-packages (from seaborn) (2.3.3)
Requirement already satisfied: matplotlib<3.6.1,>=3.4 in c:\users\danush prana\anaconda3\lib\site-packages (from seaborn) (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\danush prana\anaconda3\lib\site-packages (from matplotlib<3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\danush prana\anaconda3\lib\site-packages (from matplotlib<3.6.1,>=3.4->seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\danush prana\anaconda3\lib\site-packages (from matplotlib<3.6.1,>=3.4->seaborn) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\danush prana\anaconda3\lib\site-packages (from matplotlib<3.6.1,>=3.4->seaborn) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\danush prana\anaconda3\lib\site-packages (from matplotlib<3.6.1,>=3.4->seaborn) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\danush prana\anaconda3\lib\site-packages (from matplotlib<3.6.1,>=3.4->seaborn) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\danush prana\anaconda3\lib\site-packages (from matplotlib<3.6.1,>=3.4->seaborn) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\danush prana\anaconda3\lib\site-packages (from matplotlib<3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\danush prana\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\danush prana\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\danush prana\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib<3.6.1,>=3.4->seaborn) (1.17.0)

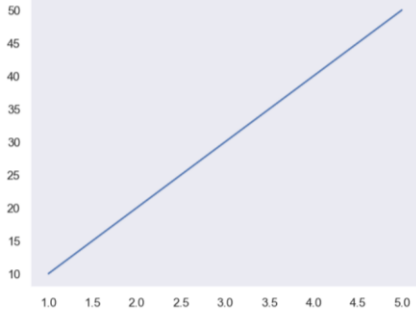
[19]: import seaborn as sns
import matplotlib.pyplot as plt
xs=[1,2,3,4,5]
ys=[10,20,30,40,50]
#theme in seaborn
#theme control the appearance of charts
#default theme
sns.set_theme()
#dark grid theme
```

Jupyter Data Visualization DAY-5 Last Checkpoint: 22 hours ago

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JupyterLab Python [conda env:base] * ○ □ ≡

```
sns.set_style("dark")
sns.set_style("white")#can u this also but know this is not executed want execute separately
#Available styles
#darkgrid
#whitegrid
sns.lineplot(xsx,ysy)
#here in seaborn we use line plot whereas in matplotlib we use plot
plt.show()
```

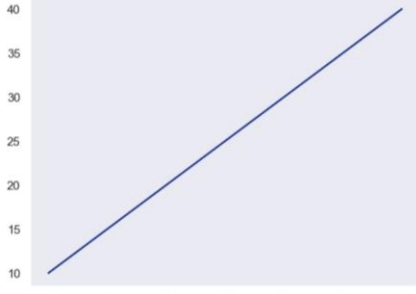


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JupyterLab Python [conda env:base] * ○ □ ≡

```
[28]: #color palettes
#A palette is a collection of colors
#view current palette
#available colors/style dark,colorblind,deep, muted,bright,pastel,
sns.set_palette("dark")
sns.lineplot(
    xs=[1,2,3,4],
    ys=[10,20,30,40]
)
plt.show()
```



Jupyter Data Visualization DAY-5 Last Checkpoint: 22 hours ago

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JupyterLab Python [conda env:base]

```
[26]: #built in dataset support
#one of seaborn's biggest advantages is built-in datasets
#view available datasets
import seaborn as sns
print(sns.get_dataset_names())

['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes', 'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri', 'geyser', 'glue',
 'healthexp', 'iris', 'mpg', 'penguins', 'planets', 'sealife', 'taxis', 'tips', 'titanic']

[27]: #Load titanic dataset
import seaborn as sns
dfsns.load_dataset("titanic")
print(df.head())

   survived  pclass    sex  age  sibsp  parch    fare embarked  class \
0         0       3   male  22.0     1     0   7.2500         S   Third
1         1       1  female  38.0     1     0  71.2833         C   First
2         1       3  female  26.0     0     0   7.9250         S   Third
3         1       1  female  35.0     1     0  53.1000         S   First
4         0       3   male  35.0     0     0   8.0500         S   Third

   who  adult_male  deck  embark_town  alive  alone
0  man         True   NaN  Southampton    no   False
1 woman        False    C   Cherbourg   yes   False
2 woman        False   NaN  Southampton   yes    True
3 woman        False    C   Southampton   yes   False
4  man         True   NaN  Southampton    no    True

[30]: #Line chart
#what is a line chart?
# a line chart connects data points with lines

#when to use?
#showing trend over time
```

Jupyter Data Visualization DAY-5 Last Checkpoint: 22 hours ago

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JupyterLab Python [conda env:base]

```
import matplotlib.pyplot as plt
months=['Jan','Feb','Mar','Apr']
sales=[100,150,200,250]
plt.plot(months,sales,markers='o')
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```



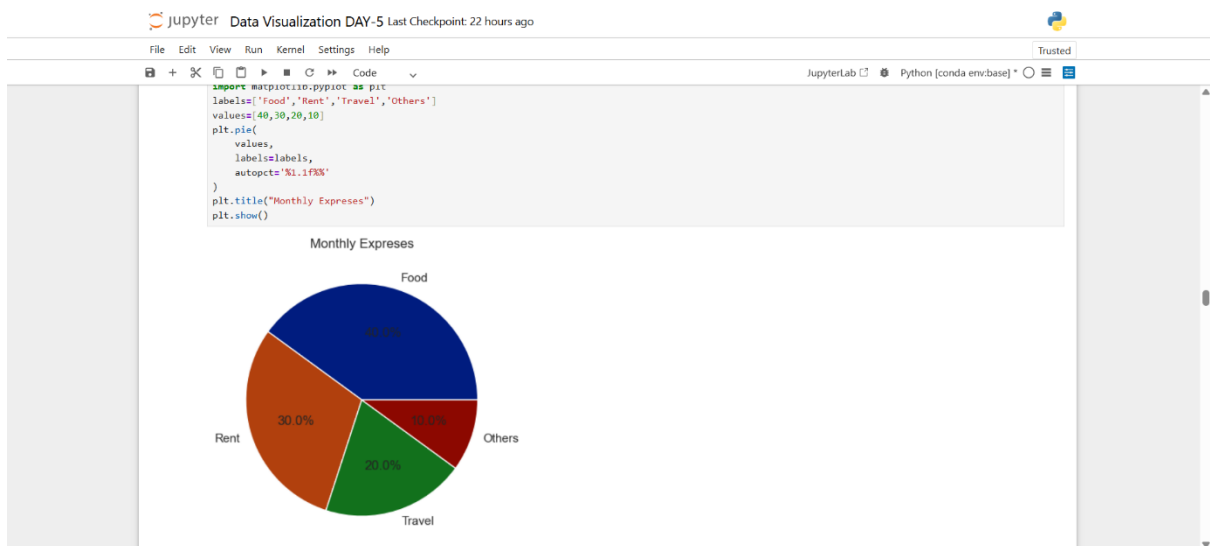
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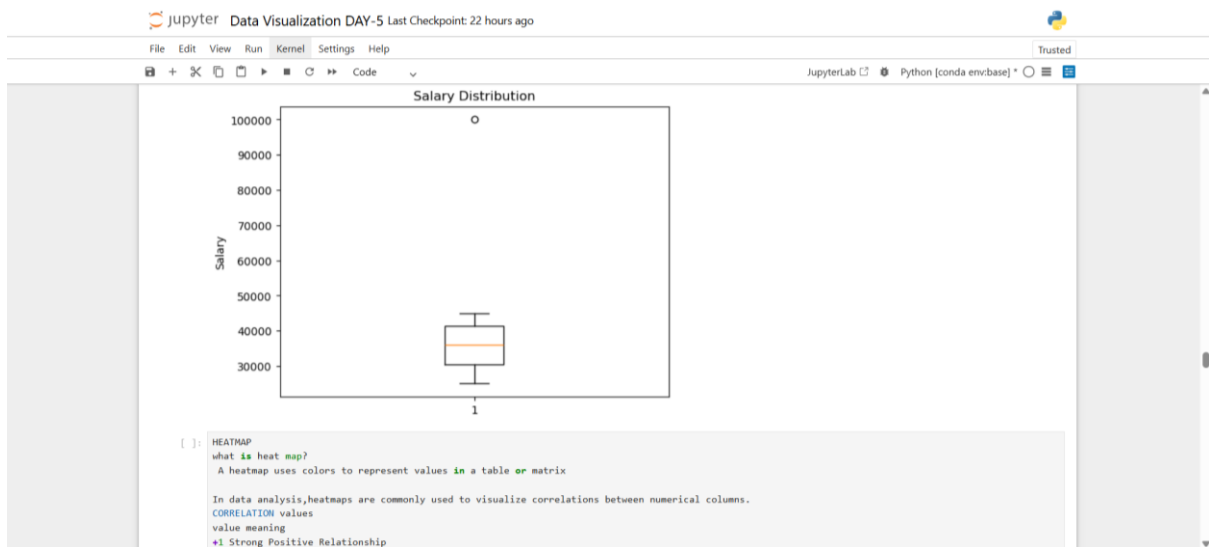
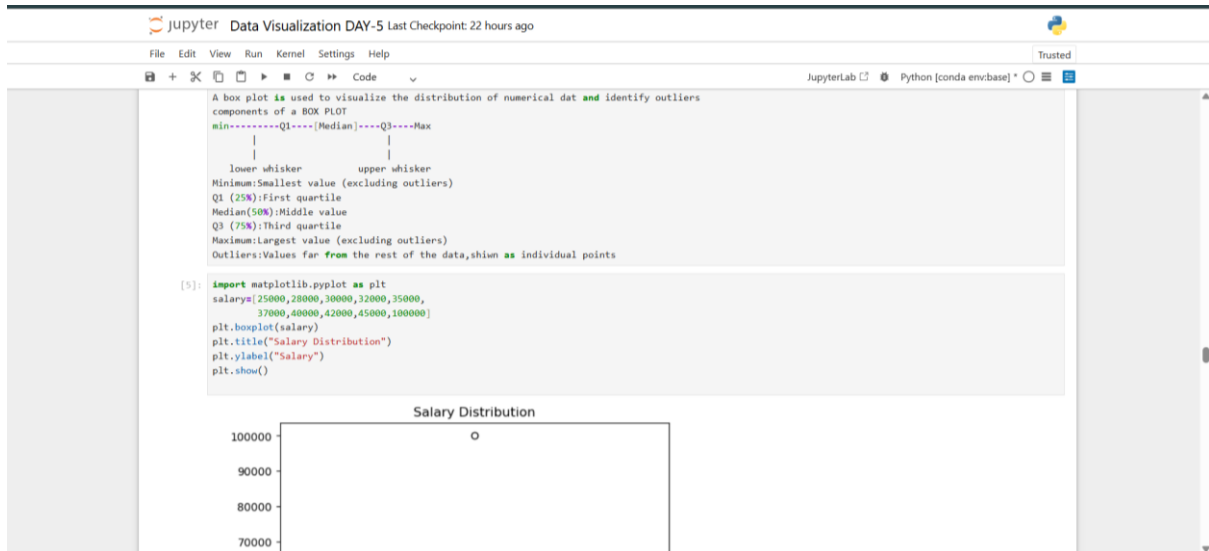
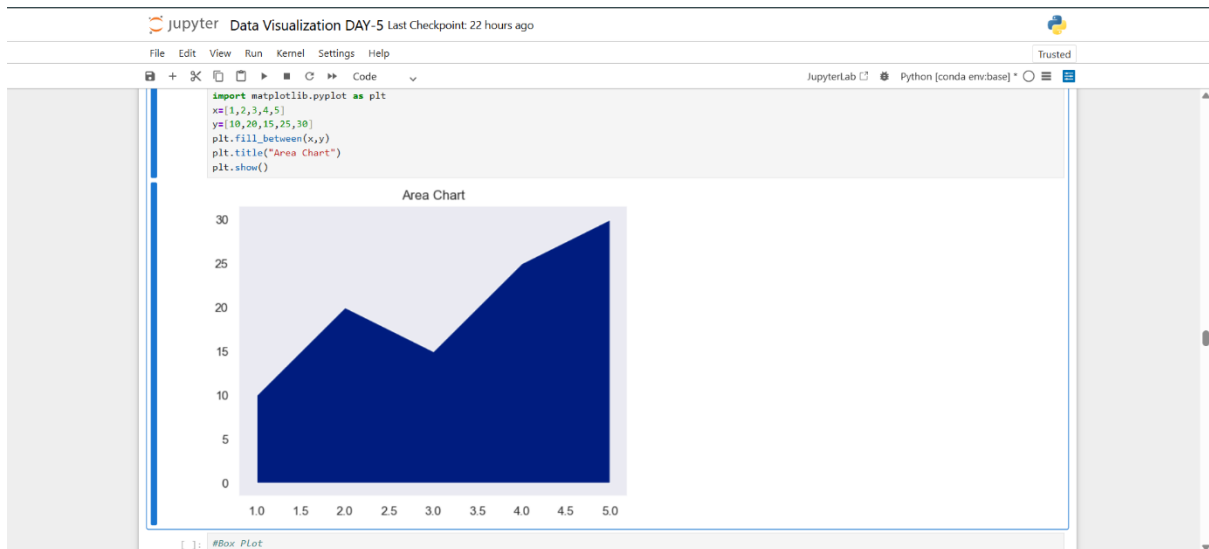
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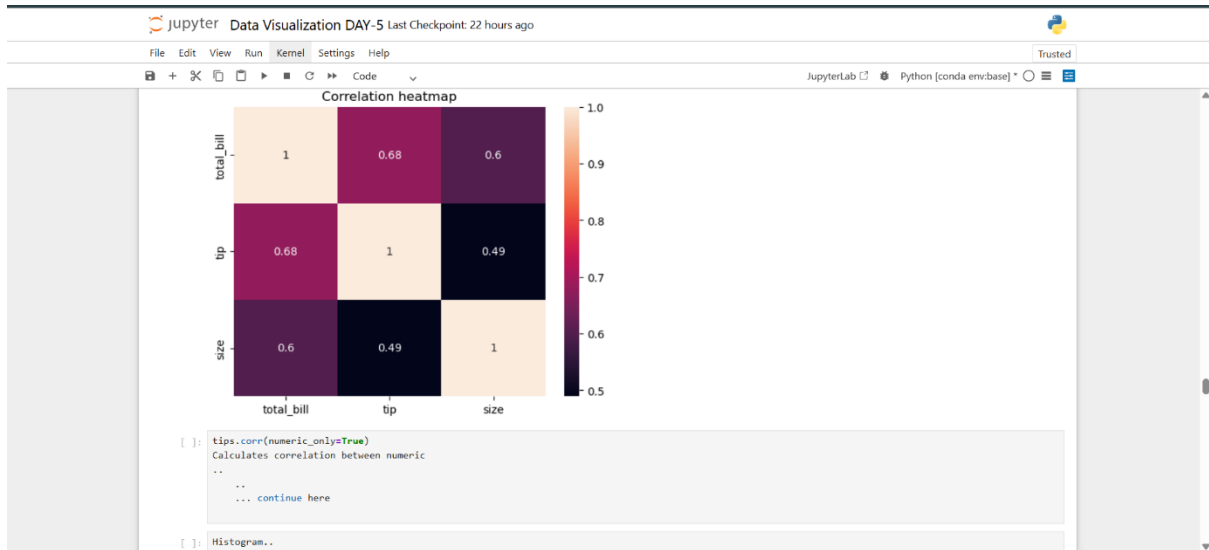
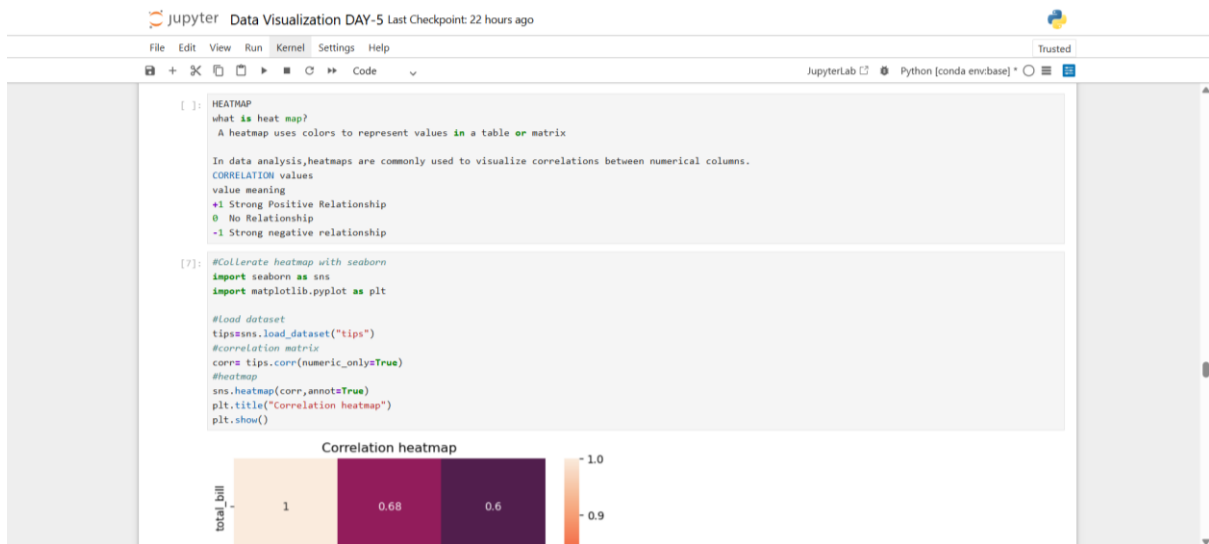
JupyterLab Python [conda env:base]

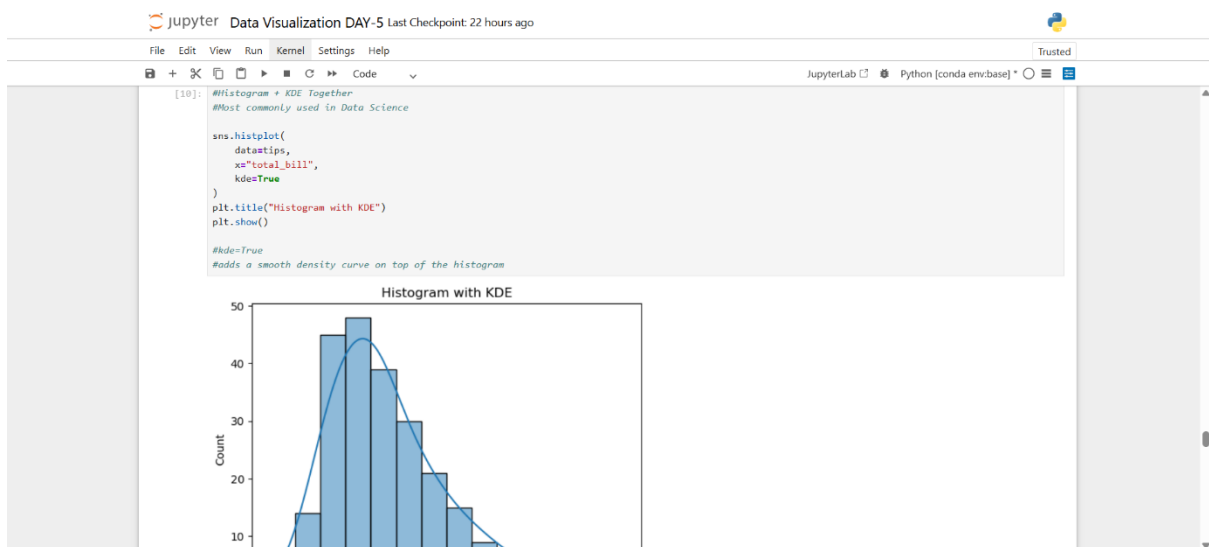
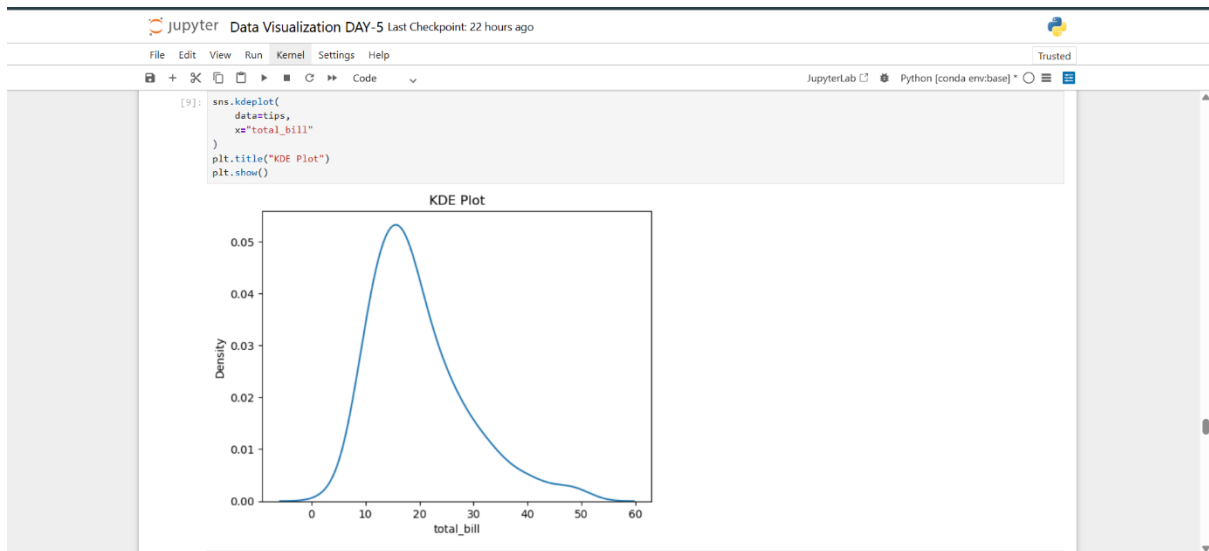
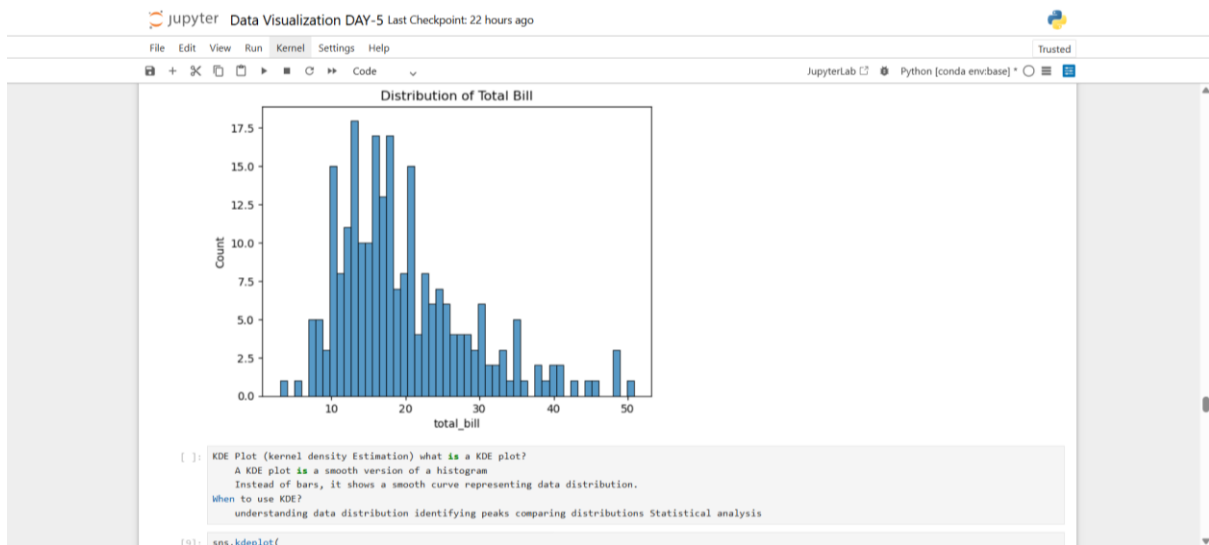
```
[36]: #Bar chart--- used for comparing
# a bar chart uses rectangular bars to compare values
#-Student marks
#-Population by city
#
#
import matplotlib.pyplot as plt
depts=['HR','IT','Sales']
employees=[20,50,35]
plt.bar(dept,employees)
plt.title("Employees by Department")
plt.xlabel("Department")
plt.ylabel("Count")
plt.show()
```

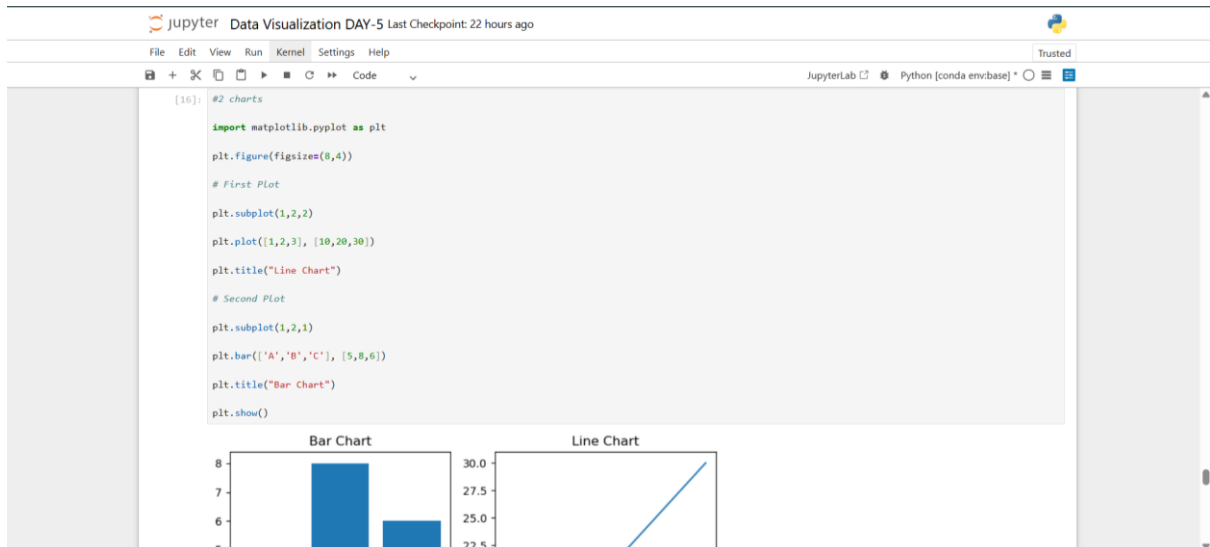
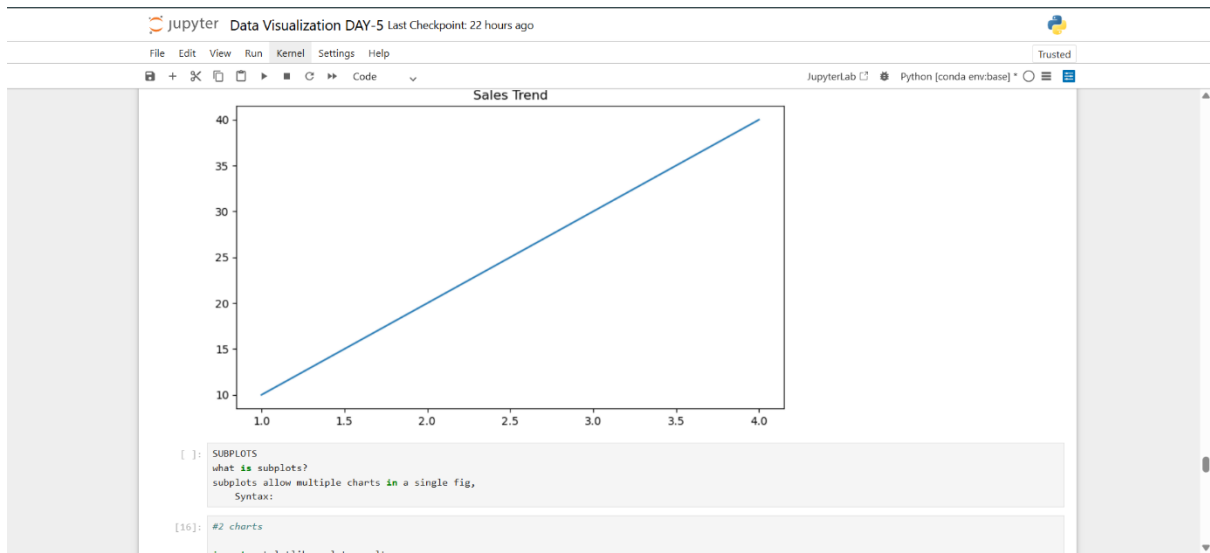
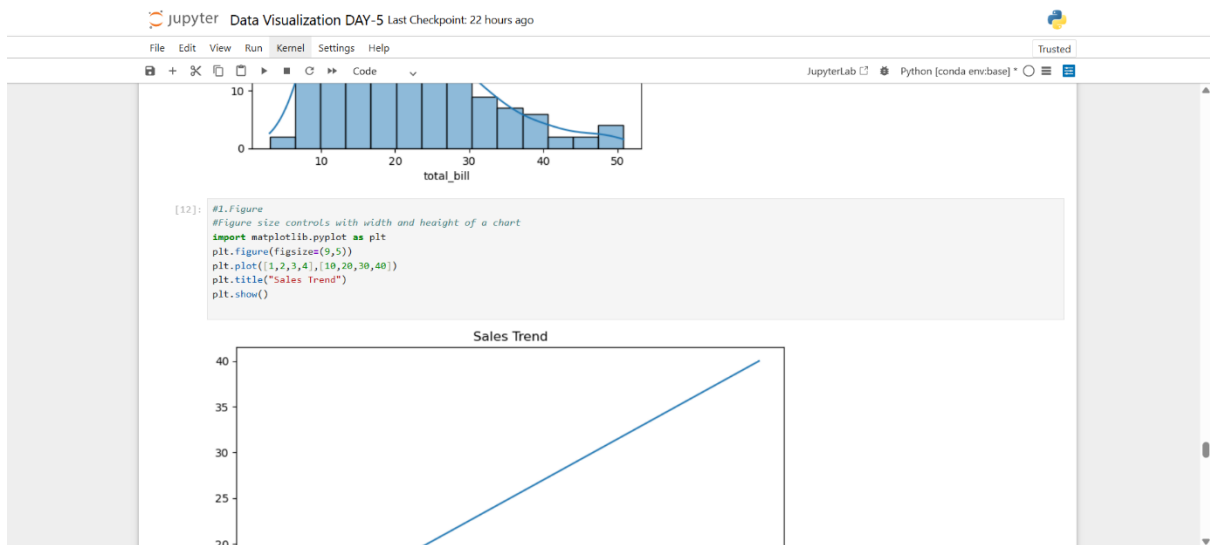


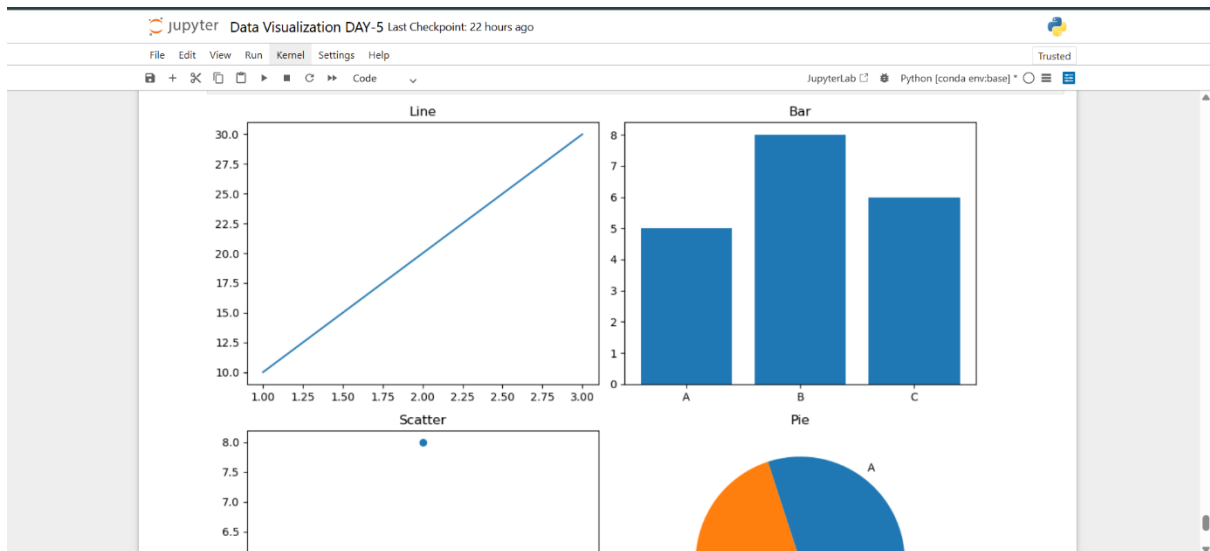
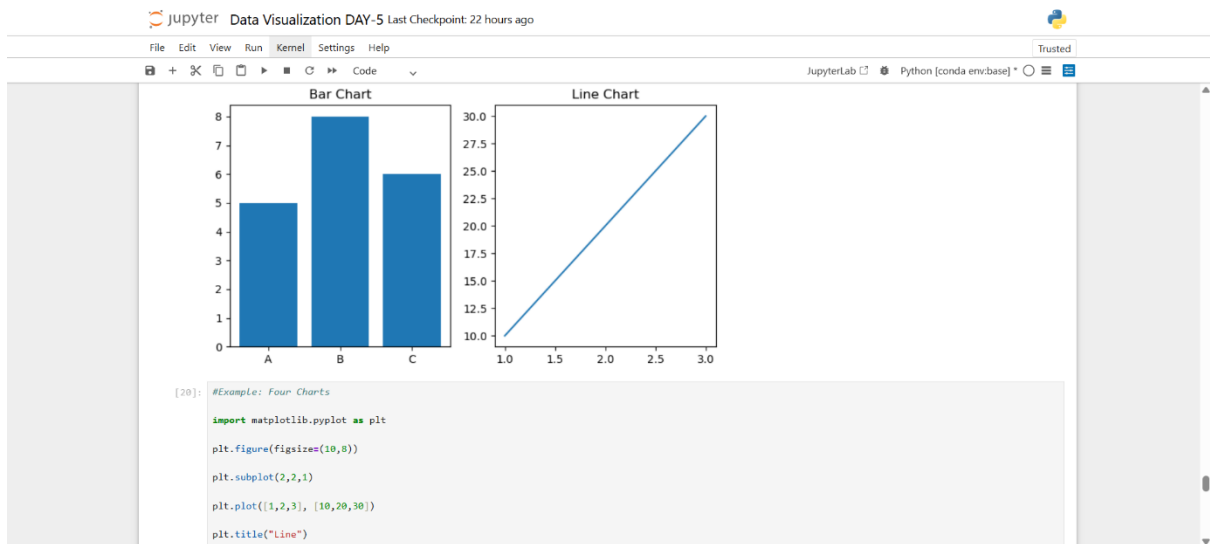


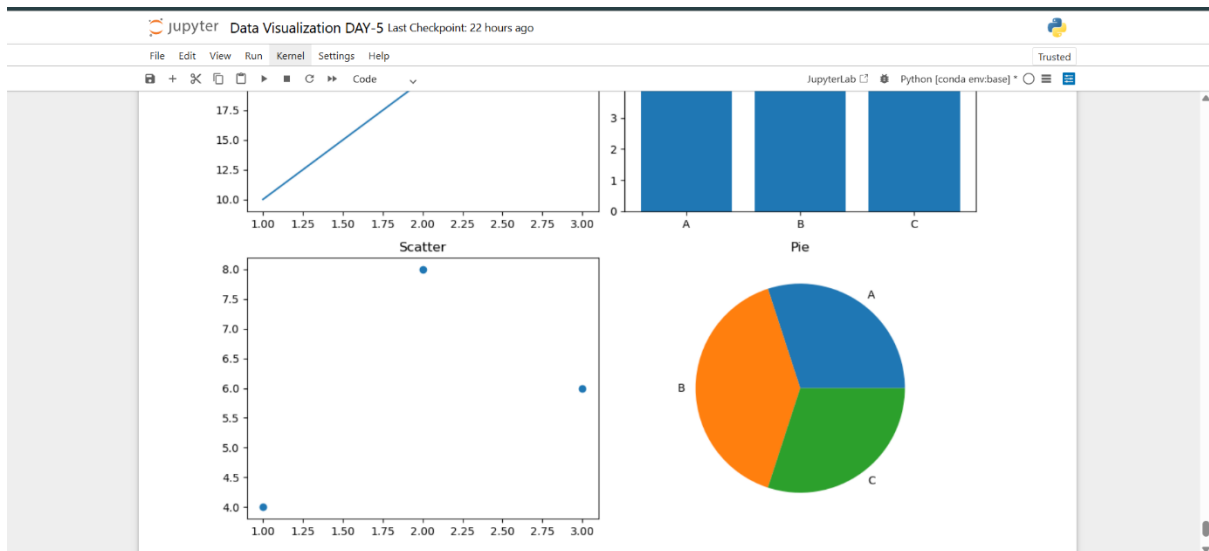




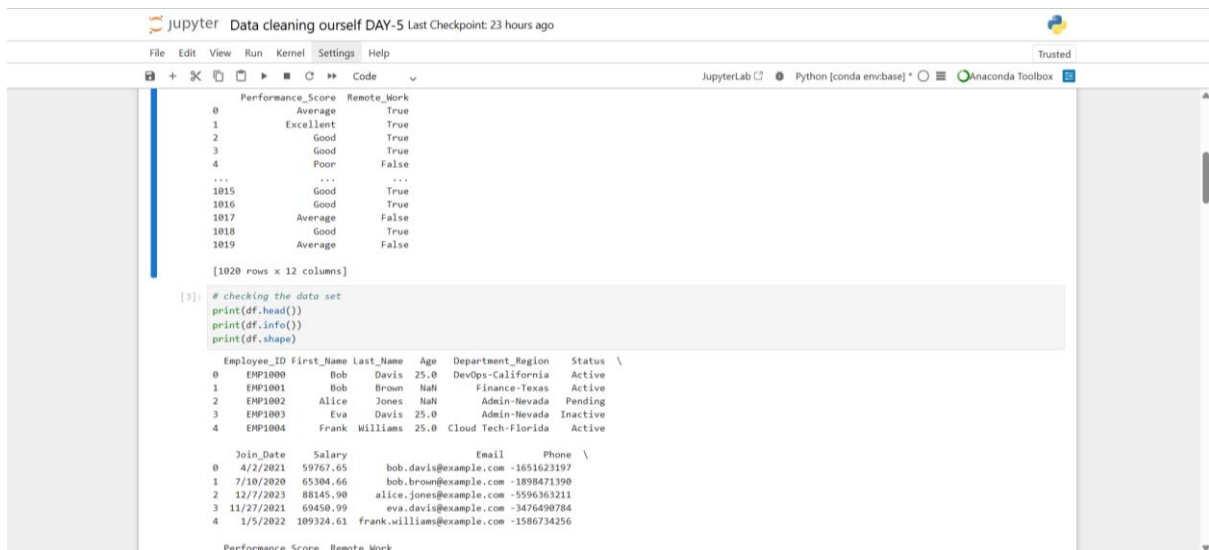
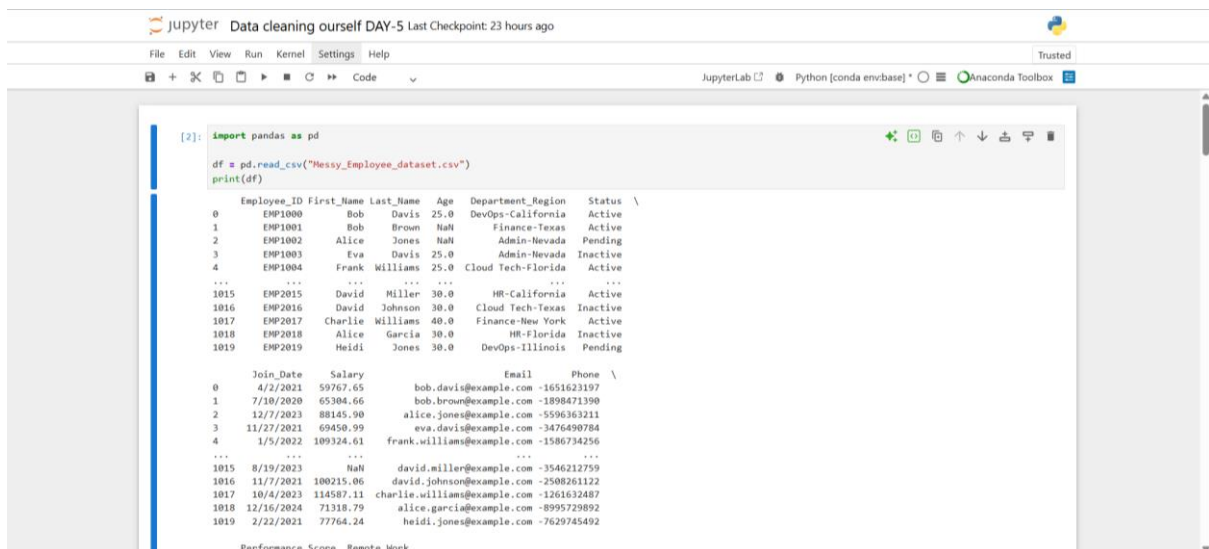








DATA CLEANING WITH A DATASET



```
jupyter Data cleaning ourself DAY-5 Last checkpoint: 23 hours ago
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JupyterLab Python [conda env:base] Anaconda Toolbox

Performance_Score Remote_Work
0 Average True
1 Excellent True
2 Good True
3 Good True
4 Poor False
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1020 entries, 0 to 1019
Data columns (total 12 columns):
 # Column Non-Null Count Dtype
---
0 Employee_ID 1020 non-null object
1 First_Name 1020 non-null object
2 Last_Name 1020 non-null object
3 Age 809 non-null float64
4 Department_Region 1020 non-null object
5 Status 1020 non-null object
6 Join_Date 1020 non-null object
7 Salary 996 non-null float64
8 Email 1020 non-null object
9 Phone 1020 non-null int64
10 Performance_Score 1020 non-null object
11 Remote_Work 1020 non-null bool
dtypes: bool(1), float64(2), int64(1), object(8)
memory usage: 88.8+ KB
None
(1020, 12)

[4]: #handles missing values
print(df.isnull().sum())

Employee_ID 0
First_Name 0
Last_Name 0
```

```
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JupyterLab Python [conda env:base] Anaconda Toolbox

Age 211
Department_Region 0
Status 0
Join_Date 0
Salary 24
Email 0
Phone 0
Performance_Score 0
Remote_Work 0
dtype: int64

[6]: #Fill numerical columns with median
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Salary"] = df["Salary"].fillna(df["Salary"].median())

[7]: ## Check current column names
print(df.columns)

# Rename columns
df.rename(columns={
    "Employee_ID": "Emp_ID",
    "Department_Region": "Department"
}, inplace=True)

print(df.columns)

Index(['Employee_ID', 'First_Name', 'Last_Name', 'Age', 'Department_Region',
       'Status', 'Join_Date', 'Salary', 'Email', 'Phone', 'Performance_Score',
       'Remote_Work'],
      dtype='object')
Index(['Emp_ID', 'First_Name', 'Last_Name', 'Age', 'Department', 'Status',
       'Join_Date', 'Salary', 'Email', 'Phone', 'Performance_Score',
       'Remote_Work'],
      dtype='object')
```

```
jupyter Data cleaning ourself DAY-5 Last checkpoint: 23 hours ago
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JupyterLab Python [conda env:base] Anaconda Toolbox

[8]: #GroupBy Department and Find Average Salary
avg_salary = df.groupby("Department")["Salary"].mean()

print(avg_salary)

Department
Admin-California 89564.108824
Admin-Florida 86876.526800
Admin-Illinois 92050.898621
Admin-Nevada 80641.359412
Admin-New York 78822.267727
Admin-Texas 81871.540000
Cloud Tech-California 83877.977931
Cloud Tech-Florida 91715.487857
Cloud Tech-Illinois 85086.828750
Cloud Tech-Nevada 85820.091000
Cloud Tech-New York 78086.093333
Cloud Tech-Texas 84466.728276
DevOps-California 82772.097143
DevOps-Florida 87116.325294
DevOps-Illinois 85678.447273
DevOps-Nevada 90001.822593
DevOps-New York 84959.466667
DevOps-Texas 86466.995185
Finance-California 87419.617188
Finance-Florida 83098.247917
Finance-Illinois 75787.702424
Finance-Nevada 76398.523448
Finance-New York 89159.340435
Finance-Texas 83832.979310
HR-California 87745.022692
HR-Florida 80460.818293
HR-Illinois 87787.819167
HR-Nevada 85849.801250
HR-New York 86366.830303
```

```
jupyter Data cleaning ourself DAY-5 Last Checkpoint: 23 hours ago

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JupyterLab Python [conda env:base] Anaconda Toolbox

Sales-California 88533.919677
Sales-Florida 90582.736667
Sales-Illinois 86982.183667
Sales-Nevada 84886.507143
Sales-New York 83930.126154
Sales-Texas 85526.279565
Name: Salary, dtype: float64

[9]: #checks duplicates
print(df.duplicated().sum())
0

[10]: #removes duplicates
df.drop_duplicates(inplace=True)

[12]: #remove duplicate value column-wise from emp id and email
print(df["Emp_ID"].duplicated().sum())

df.drop_duplicates(subsets="Emp_ID", inplace=True)
df.drop_duplicates(subsets="Email", inplace=True)
0

[13]: #check current data type
print(df.dtypes)

Emp_ID      object
First_Name  object
Last_Name   object
Age         float64
Department  object
Status      object
Join_Date   object
Salary      float64
dtype: object
```

```
jupyter Data cleaning ourself DAY-5 Last Checkpoint: 23 hours ago

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JupyterLab Python [conda env:base] Anaconda Toolbox

Salary      float64
Email       object
Phone       int64
Performance_Score  object
Remote_Work bool
dtype: object

[14]: #Convert columns
df["Age"] = df["Age"].astype(int)

df["Salary"] = df["Salary"].astype(float)

df["Join_Date"] = pd.to_datetime(df["Join_Date"])

[17]: #string operation removes extra space
df["First_Name"] = df["First_Name"].str.strip()
df["Last_Name"] = df["Last_Name"].str.strip()

#converts to title case
df["First_Name"] = df["First_Name"].str.title()
df["Last_Name"] = df["Last_Name"].str.title()

#replace values
df["Department"] = df["Department"].str.replace(
    "CSE",
    "Computer Science"
)

[19]: #final check
print(df.info())

print(df.head())

print(df.duplicated().sum())
```

```
jupyter Data cleaning ourself DAY-5 Last Checkpoint: 23 hours ago

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JupyterLab Python [conda env:base] Anaconda Toolbox

print(df.info())

print(df.head())

print(df.duplicated().sum())

<class 'pandas.core.frame.DataFrame'>
Int64Index: 64 entries, 0 to 248
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  --
 0   Emp_ID                64 non-null    object
 1   First_Name            64 non-null    object
 2   Last_Name             64 non-null    object
 3   Age                   64 non-null    int64
 4   Department            64 non-null    object
 5   Status                64 non-null    object
 6   Join_Date             64 non-null    datetime64[ns]
 7   Salary                64 non-null    float64
 8   Email                 64 non-null    object
 9   Phone                 64 non-null    int64
10  Performance_Score     64 non-null    object
11  Remote_Work           64 non-null    bool
dtypes: bool(1), datetime64[ns](1), float64(1), int64(2), object(7)
memory usage: 6.1+ KB
None
   Emp_ID First_Name Last_Name Age  Department  Status Join_Date \
0  EMP1000      Bob    Davis   25  DevOps-California  Active  2021-04-02
1  EMP1001      Bob    Brown   30    Finance-Texas  Active  2020-07-10
2  EMP1002    Alice    Jones   30  Admin-Nevada  Pending  2023-12-07
3  EMP1003      Eva    Davis   25  Admin-Nevada  Inactive  2021-11-27
4  EMP1004    Frank  Williams   25  Cloud Tech-Florida  Active  2022-01-05

   Salary      Email      Phone  Performance_Score \
0  88533.919677  Bob.Davis@company.com  90582736667  88533.919677
1  90582.736667  Bob.Brown@company.com  86982183667  90582.736667
2  86982.183667  Alice.Jones@company.com  84886507143  86982.183667
3  84886.507143  Eva.Davis@company.com  83930126154  84886.507143
4  83930.126154  Frank.Williams@company.com  85526279565  83930.126154
```

jupyter

Data cleaning ourself DAY-5 Last Checkpoint: 23 hours ago

File

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Kernel

Settings

Help

Trusted

Code

JupyterLab

Python [conda env:base] *

Anaconda Toolbox

	EMPLOYEE	Age	Experience	Start Date	
0	EMP1000	Bob	Davis	25	DevOps-California
1	EMP1001	Bob	Brown	30	Finance-Texas
2	EMP1002	Alice	Jones	30	Admin-Nevada
3	EMP1003	Eva	Davis	25	Admin-Nevada
4	EMP1004	Frank	Williams	25	Cloud Tech-Florida

	Salary	Email	Phone	Performance_Score	\
0	59767.65	bob.davis@example.com	-1651623197	Average	
1	65304.66	bob.brown@example.com	-1898471390	Excellent	
2	88145.90	alice.jones@example.com	-5596363211	Good	
3	69450.99	eva.davis@example.com	-3476490784	Good	
4	109324.61	frank.williams@example.com	-1586734256	Poor	

	Remote_Work
0	True
1	True
2	True
3	True
4	False
0	

[20]:

#save the cleaned data
df.to_csv("Cleaned_Employee_Dataset.csv", index=False)

[]: